

PEDRO AUGUSTO MAGALHÃES

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Education

University of São Paulo

Bachelor's Degree in Mechanical Engineering with emphasis on Computing

Mar. 2020 – Present

São Paulo, Brazil

Instituto Superior Técnico

Double Degree in Mechanical Engineering with focus on Control Systems

Sep. 2023 – May 2025

Lisbon, Portugal

Experience

Andritz Hydropower

Systems Engineering Intern

Oct. 2025 – Present

- Project management of generators and compensators, from material tracking for manufacturing to factory processes, ensuring on-time project delivery.
- Development of VBA models in Excel to support project management, creating a platform for the department.

Instituto Superior Técnico

Researcher

Nov. 2023 – Dec. 2024

- Research in Structural and Computational Mechanics, focusing on optimization techniques and passive control applied to structural vibration reduction. Development of a generic ANSYS model to validate multiple configurations, converting static simulations into dynamic ones, increasing efficiency and significantly reducing computational time.
- Multimodal optimization of piezoelectric patches number and placement for effective structural vibration reduction

University of São Paulo

Researcher

Aug. 2022 – Aug. 2023

- Research in Control Systems aimed at optimizing operational performance of offshore oil drilling platforms. Development of a comparative model between Drill Control techniques, incorporating stochastic uncertainties and nonlinearities in the bit-rock interaction.
- Robustness analysis of oil well drilling control techniques under uncertain nonlinear interaction between rock and bit.

EESC jr.

Technical Consultant

Aug. 2021 – Aug. 2022

- Acted as a mechanical project leader, responsible for technical communication with clients, scope definition, and coordination of the development team. Developed customized mechanical solutions, including vending machines, wire bending machines, automated stock replenishment systems, and technical feasibility studies.

University of São Paulo

Programming Teaching Assistant

Jan. 2021 – Jul. 2021

- Teaching assistant for the Introduction to Programming course for Electrical Engineering at EESC-USP.

Technical Skills

Development: Python, SQL, C++, MATLAB, APDL, VBA

Software: ANSYS, GMesh, OpenFOAM, ParaView, SolidEdge, SolidWorks, PowerBI, Excel

Certifications / Additional Activities

- Offshore Engineering for Renewable Energy (90h) – Delft, Netherlands
Authentication code: M29K-MNEY-EFA9-M3RA
- Efficiency Evaluation of Turbomachinery (90h) – Vienna, Austria
Authentication code: DAAC-JCWJ-V73U-ZXG4
- Sustainable Hydropower Development (90h) – Trondheim, Norway
Authentication code: 9662-3BPB-YMFQ-2WFFV
- 2nd Place at TecStorm 24/25 and 4th Place at TecStorm 23/24 (University Hackathon) – Lisbon, Portugal
- Staff Member at ECCOMAS International Conference – Lisbon, Portugal
- Participant and author of a paper presented at the International Conference on Numerical Uncertainty – Fortaleza, Brazil

Languages

Portuguese: Native

English: Proficient